

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DRK-006 - Dirok-006

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000068
Field / District	Dirok / Dibrugarh
Block	OALP-III-08
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2007-09-24 / 2008-01-06
Measured / TVD	2139 m / 2108 m
Trajectory	Directional; inc 18.3 deg; az 89.7 deg
Formation Interval	1201-1724 m
Lithology	Argillaceous sandstone and shale
Pore Pressure Reference	2311 psi
Fracture Gradient	14.69 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Low (39.2/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	534	Surface
9 5/8 in casing	1240	Intermediate
7 in liner/casing	1882	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2007-12-08 1596 m	Torque Spike	Critical	Torque rose without corresponding pressure anomaly. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2007-11-10 1077 m	Torque Spike	High	Torque rose without corresponding pressure anomaly. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2007-11-17 1209 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2007-11-26 1379 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2007-10-19 639 m	Pressure Spike	Medium	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Correlate pressure anomalies with flow before escalating risk.
2007-11-18 1229 m	Stuck Pipe	Medium	Rotary movement deteriorated progressively. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2007-12-02 1488 m	Torque Spike	Medium	Torque rose without corresponding pressure anomaly. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.

Completion Summary

The synthetic well reached 2139 m MD in the Girujan interval. The NWIS knowledge record links 10 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.